THE COMPOSITE OF THE GYSIN-AT-INFINITY CLASS WITH THE PUSHFORWARD FROM INFINITY EQUALS THE GYSIN PULLBACK AT INFINITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FAR (sources/stacks-chow/pinned-original-source.tex line 10132).

Allowed standard prerequisites (editorial).

- The Gysin-at-infinity class and its defining property. Same chapter, Lemma labelled lemma-gysin-at-infty (source line 10026), invoked in the statement. Supplies the class C and the characterisation C cap alpha = i_infinity^* beta which is what the proof must match.
- The two rationally equivalent effective Cartier divisors of the projective line give the same cycle class. Same chapter, Lemma labelled lemma-support-cap-effective-Cartier (source line 5138), invoked at proof lines 10142-10143. Supplies that i_{infinity,*}i_infinity^* beta and i_{0,*}i_0^* beta represent the same class on W.
- The line bundle lemmas identifying fibre restrictions after pullback along the zero section. Same chapter, Lemmas labelled lemma-linebundle (source line 5737) and lemma-linebundle-formulae (source line 5775), invoked at proof lines 10148-10149. Supply that the restriction of beta to b^{-1}(A^1_X) restricts to the flat pullback of i_0^* beta, which may be verified after pullback along i_0.

Source reference. Original lemma 0F9J: sources/stacks-chow/pinned-original-source.tex, source lines 10026–10026. Same-chapter pointer into the author environment lemma at source line 10026; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02T9: sources/stacks-chow/pinned-original-source.tex, source lines 5138–5138. Same-chapter pointer into the author environment lemma at source line 5138; invoked at proof line 10143 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 0B74: sources/stacks-chow/pinned-original-source.tex, source lines 5737–5737. Same-chapter pointer into the author environment lemma at source line 5737; invoked at proof line 10149 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F96: sources/stacks-chow/pinned-original-source.tex, source lines 5775–5775. Same-chapter pointer into the author environment lemma at source line 5775; invoked at proof line 10149 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Lemma 0F9J we have* $C \circ (W_\infty \to X)_* \circ i_\infty^* = i_\infty^*$.

Proof. Let $\beta \in \mathrm{CH}_{k+1}(W)$. Denote $i_0 : X = X \times \{0\} \to W$ the closed immersion of the fibre over 0 in $\mathbf{P}^1$. Then $(W_\infty \to X)_* i_\infty^* \beta = i_0^* \beta$ in $\mathrm{CH}_k(X)$ because $i_{\infty,*} i_\infty^* \beta$ and $i_{0,*} i_0^* \beta$ represent the same class on W (for example by Lemma 02T9) and hence pushforward to the same class on X. The restriction of β to $b^{-1}(\mathbf{A}^1_X)$ restricts to the flat pullback of $i_0^* \beta = (W_\infty \to X)_* i_\infty^* \beta$ because we can check this after pullback by i_0, see Lemmas 0B74 and 0F96. Hence we may use β when computing the image of $(W_\infty \to X)_* i_\infty^* \beta$ under C and we get the desired result. $\square$